MATH 526 Section 001 Lab 2 Fall 2004

Part I

In this part we learn to solve a lower triangular system by the method of forward substitution.

Create the following lower triangular matrix **A** and the vector **b**.

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 4 & 5 & 2 & 0 \\ 3 & -2 & 1 & 1 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ 6 \\ 4 \end{bmatrix}$$

The dimension of a matrix can be determined by the command **size**. Try the following.

>>**[m,n]=size(A)**
>>**[m,n]=size(b)**

Now we solve **Ax=b** by forward substitution.

>>**x(1)=b(1)/A(1,1)**
>>**x(2)=(b(2)–A(2,1)∗x(1))/A(2,2)**
>>**x(3)=(b(3)–A(3,1)∗x(1)–A(3,2)∗x(2))/A(3,3)**
>>**x(4)=(b(4)–A(4,1)∗x(1)–A(4,2)∗x(2)–A(4,3)∗x(3))/A(4,4)**

The solution is now represented by a row vector. We can of course also represent the solution by a column vector

>>**x=x′**
>>**clear x**

This command removes the vector **x**.

Note that a product such as **A(3,1)∗x(1)+A(3,2)∗x(2)** can be expressed as the product of the vectors

$$[A(3,1)\ \ A(3,2)] \quad \text{and} \quad \begin{bmatrix} x(1) \\ x(2) \end{bmatrix}.$$

Hence we can replace **A(3,1)∗x(1)+A(3,2)∗x(2)** by **A(3,1:2)∗x(1:2)′**. (Make sure you understand the meaning of all the terms. You saw them last week!)

We can therefore simplify the commands as follows.

>>x(1)=b(1)/A(1,1)
>>x(2)=(b(2)–A(2,1:1)∗x(1:1)′)/A(2,2)
>>x(3)=(b(3)–A(3,1:2)∗x(1:2)′)/A(3,3)
>>x(4)=(b(4)–A(4,1:3)∗x(1:3)′)/A(4,4)

Of course we can simplify further by using a simple **for** loop as follows. (Use the command **help for** for more information about the **for** loop.)

>>clear x
>>x(1)=b(1)/A(1,1);
>>for i=2:1:4
x(i)=(b(i)–A(i,1:i–1)∗x(1:i–1)′)/A(i,i);
end

We have suppress the printing of the results of the computation. To see the answer, type

>>x=x′

We notice that the commands we used to solve **Ax=b** are general commands, i.e., they can be applied to solve any 4×4 lower triangular system. So instead of typing the same commands every time we want to solve a 4×4 lower triangular system, we can create a file containing these commands and instruct MATLAB to execute the commands in the file.

Pop up an M-file window and type in the following lines.

```
function x=forward(A,b)
x(1)=b(1)/A(1,1);
for i=2:1:4
x(i)=(b(i)–A(i,1:i–1)∗x(1:i–1)′)/A(i,i);
end
x=x′;
```

Save this file as forward.m   Go back to your MATLAB window and type the following commands.

>>clear x
>>x=forward(A,b)

Matlab will execute the file forward.m using the matrix **A** and the vector **b** you have created before as the inputs and return the solution vector **x** as the output.

We can also write an M-file that will solve lower triangular systems of any dimension by using the command **size**. Modify your forward.m file as follows.

```
function x=forward(A,b)
[m,n]=size(A);
if m~=n, error('The input matrix is not a square matrix.'), end
x(1)=b(1)/A(1,1);
for i=2:1:m
x(i)=(b(i)-A(i,1:i-1)*x(1:i-1)')/A(i,i);
end
x=x';
```

Now you can solve any lower triangular system by using the M-file forward.m .

Question: What does the third line of forward.m do?

Let us create a 3×4 matrix **B**:

>>B=A(1:3,1:4)

Now we input **B** and **b** into forward.m:

>>x=forward(B,b)

What happened?

Use **help error** to find out more about the **error** command. Can you modify the M-file forward.m further so that it can also detect any discrepancy between the dimensions of A and b?

Before you start working on Part II, clear all the variables in your MATLAB window.